DAMA: A Dual Alignment Framework for Enhanced LLM-Powered Recommendations

Qingwen Zeng[1*], Lining Chen[1*], Jushang Qiu[2], Fangchen Liu[1], Huaming Chen[1**], and Ling Chen[3]
[1]The University of Sydney, Sydney, Australia
Email: {qzen5227, lche8972, fliu6119}@uni.sydney.edu.au, huaming.chen@sydney.edu.au
[2]Australian National University, Canberra, Australia
Email: jushang.qiu@anu.edu.au
[3]University of Technology Sydney, Sydney, Australia
Email: Ling.Chen@uts.edu.au

2025 IEEE International Conference on Data Mining (ICDM) | 979-8-3315-9599-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDM65498.2025.00182

***Abstract*—Recommender systems play a pivotal role in personalized content delivery but still face significant challenges in sparse interaction scenarios such as cold start conditions. Recent works have explored the integration of large language models (LLMs) into recommendation tasks. It has shown early promise but also reveals some limitations, such as hallucinations and overconfident outputs due to a lack of targeted modeling process. This paper introduces DAMA, a novel framework for Dual Alignment for Motivation-Aware recommendation. DAMA is the first framework to systematically incorporate LLM alignment into recommender system modeling. It employs a dual alignment mechanism that jointly optimizes semantic representations from two dimensions: user motivation modeling and product attribute modeling. By leveraging natural language feedback, DAMA facilitates a structured and interpretable alignment process. Extensive experimental results on multiple real-world datasets demonstrate that DAMA significantly improves recommendation performance, especially in cold start scenarios. We anticipate that DAMA will pioneer a new alignment-centric and semantically grounded paradigm for recommender systems.**

I. Introduction

Recommender systems are a core component of modern digital platforms, playing a pivotal role in information filtering and personalized content delivery [1]. Ensuring the recommendation accuracy is critical. In recent years, recommendation technologies have evolved from traditional collaborative filtering methods to sequential recommendation models capable of capturing user behavior patterns over time [2], [3]. While these methods perform well in scenarios with abundant interaction data, they often struggle in cold start scenarios, where new users or items lack sufficient historical interaction data [4]. To address this challenge, recent studies have explored the integration of large language models (LLMs) into recommender systems with greater granularity. Leveraging their powerful semantic understanding and generative capabilities, LLMs can utilize additional textual information, i.e., product descriptions and user reviews, to generate rich semantic representations for new users or items [5]. This enables improved performance in data-sparse environments, positioning LLMs as a promising solution for cold start scenarios [6].

However, the application of LLMs in recommendation scenarios presents several limitations, particularly hallucination and overconfidence. Hallucination refers to the generation of false or distorted semantic content by the model in the absence of sufficient evidence [7], while overconfidence describes the model's tendency to produce highly certain outputs even when its internal confidence is low [8]. These issues can distort user intent interpretations or item attributes representations, resulting in misaligned user preferences and inconsistent recommendations. It will directly undermine the effectiveness and interpretability of recommender systems [5].

These limitations are rooted in the prevailing "textual reasoning–semantic induction" paradigm. Researchers typically rely on LLMs to infer users' deep motivations and preferences from interaction histories, textual content, and product descriptions, with the goal of deriving representations that generalize well [9]. However, in recommendation scenarios, such motivational inference often lacks complete contextual information or explicit supervision, making LLMs prone to generating seemingly coherent yet unverified conclusions in the face of insufficient evidence (Hallucination) [7], [10]. At the same time, the model's probabilistic outputs do not always align with actual prediction accuracy, frequently exhibiting high-confidence predictions on low-confidence instances (Overconfidence). When these biased inferences are adopted by downstream modules, they propagate through the recommendation pipeline, becoming increasingly amplified [5]. This cascading effect ultimately undermines the overall performance and trustworthiness of the recommendation system.

To harness the powerful semantic modeling capabilities of LLMs while mitigating the inferential bias, this paper proposes a novel Dual Alignment for Motivation-Aware recommendation framework DAMA that, for the first time, places LLM alignment at the core of recommendation modeling. Aimed at addressing the prevalent issues of hallucination and overconfidence in recommendation contexts, DAMA establishes a user preference modeling paradigm driven by semantic alignment.

Specifically, on the user side, DAMA incorporates users' explicit feedback—such as authentic reviews and behavioral signals—as alignment cues to guide the LLM in more ac-

[*] These authors contributed equally.
[**] Corresponding author.

2374-8486/25/$31.00 ©2025 IEEE
DOI 10.1109/ICDM65498.2025.00182

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

curately identifying personalized motivations and preferences. On the item side, DAMA calibrates product representations using collective user experience data, thereby enhancing the credibility of model reasoning and its ability to distinguish fine-grained semantic attributes.

Through this dual alignment mechanism, the recommendation system effectively mitigates preference misjudgments arising from hallucinations and reasoning biases. It significantly improves recommendation accuracy and robustness in semantically complex scenarios. By positioning LLM alignment as a central paradigm for modeling user preferences and motivations, this design pioneers a structured methodological foundation and provides a reasoning safety guarantee for future research in semantic recommendation. We summarize the contributions as follows:

- **Identifying latent risks in semantic inference with LLMs:** We provide a systematic analysis for LLM-powered recommendation systems, particularly focusing on the hallucination and overconfidence issues as key challenges that require explicit modeling and mitigation within semantic inference.
- **Design and implementation of the DAMA framework:** We present DAMA, a novel framework with a user-item dual alignment mechanism. DAMA leverages real-world feedback to guide LLM-based inference, thereby reducing inference biases and improving the precision of user motivation modeling and semantic matching.
- **Establishing an alignment-aware recommendation modeling paradigm:** For the first time, we elevate LLM alignment from a practical technique to a theoretical foundation within recommender system modeling. We validate DAMA' effectiveness in cold-start and semantically complex scenarios, pioneering methodological groundwork for research in alignment-centric semantic recommendation.

II. Related Work

Recent progress has seen a surge in incorporating large language models (LLMs) into recommender systems. P5 [11] reformulates various recommendation tasks into a unified text-to-text framework by encoding user, item, and task information into prompt templates. While this effectively recasts recommendation as a language modeling problem, it depends heavily on item IDs, limiting the use of LLMs' contextual reasoning and introducing biases from pretraining data. TALLRec [12] applies instruction tuning to better adapt LLMs for user-item interaction prediction, though it only uses product titles and lacks deeper modeling of behavioral semantics. LLM-REC [13] enhances item understanding via diverse prompt types, but overlooks user-side reasoning, resulting in shallow personalization. RdRec [14] extracts preferences and item attributes from reviews to finetune a smaller LLM, surfacing more realistic signals but still relying on ID-based inputs, which may cause hallucination. ToolRec [15] addresses this by treating the LLM as a proxy user that engages in multi-turn reasoning aided by external retrieval tools, thereby supporting more deliberate preference estimation. A-LLMRec [16] fuses collaborative filtering (CF) with LLMs by injecting CF-derived embeddings as soft prompts, enhancing contextual awareness in generation. Collectively, these works highlight an evolving shift from shallow LLM-driven recommendation toward more semantically grounded, user-intent-aligned paradigms.

III. Preliminary

LLM alignment aims to guide models to generate outputs that reflect genuine human intent, ensuring their behavior accords with human goals rather than relying on superficial associations or statistical correlations learned from training corpora [17]. This objective is typically achieved by fine-tuning on high-quality datasets of human feedback and preferences. In this work, our core alignment task is to guide the LLM to generate semantic reasoning that faithfully reflects users' true motivations and product experiences, thereby avoiding subjective inferences from limited or ambiguous context.

IV. Methodology

DAMA consists of three key modules as shown in Fig. 1. The first is the **Motivation-Aligned Profile Construction Module**, which extracts the user's underlying motivations and preferences. The second is the **Experience-Aligned Attribute Extraction Module**, which identifies and structures key item attributes relevant to user decision-making. The third is the **Alignment-Guided Recommendation Module**, which matches user motivations with item attributes to generate final recommendations.

A. Motivation-Aligned Profile Construction Module

1) Motivation Modeling via Role-Based Priors: The core modeling logic of this module is grounded in a Bayesian inference structure, where user identity is explicitly modeled as a prior constraint in the motivation inference process. Specifically, given a user's historical interaction sequence with products, denoted as S, we prompt a large language model (LLM) to infer the user's potential social role r (e.g., parent, student, government officer). This inferred role serves as a semantic prior that reflects typical behavioral motivations associated with specific societal identities.

We formalize the reasoning process using the following Bayesian decomposition:

$$P(m \mid S) = \sum_{r} P(m \mid r) \cdot P(r \mid S) \tag{1}$$

where S denotes the user's historical interaction sequence, r denotes the inferred social role of the user, m represents the user's underlying motivation to be inferred, $P(r \mid S)$ is the probability of assuming role r given sequence S, and $P(m \mid r)$ is the likelihood of motivation m conditioned on role r.

This formulation constructs a *semantic role attribution mechanism*, in which user motivation is not inferred directly from behavioral data, but is instead mediated through an intermediate identity prior. Such a design provides structured semantic constraints and mitigates difficulties caused by sparse

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

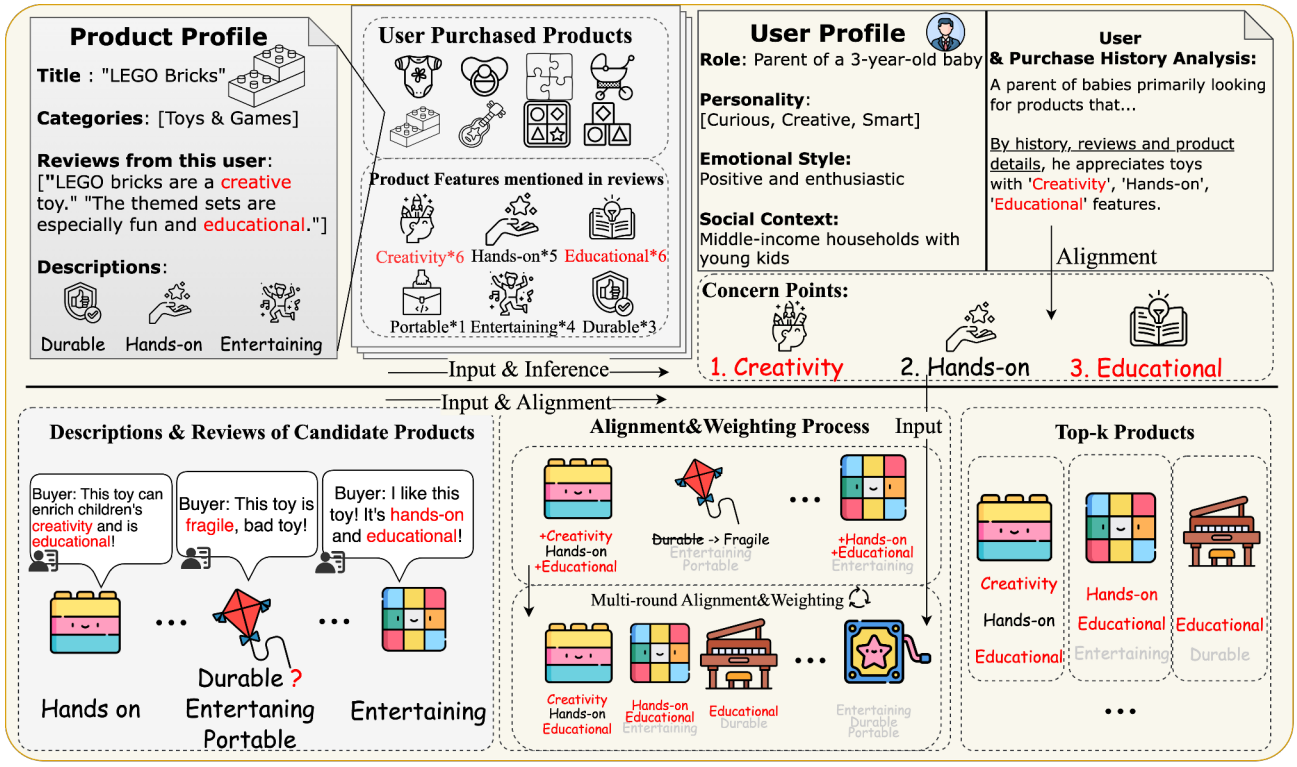


Fig. 1: This figure illustrates the model inference pipeline. The upper part depicts the Motivation-Aligned Profile Construction Module, while the lower part comprises the Experience-Aligned Attribute Extraction Module (left) and the Alignment-Guided Recommendation Module (right).Red text indicates aligned user/item features via alignment techniques; black text denotes preference features extracted from product descriptions. Grey text indicates features that do not match with the user's intent.

or noisy interaction signals. For example, both a "parent" and a "fitness enthusiast" may purchase protein powder, yet their motivations can differ significantly: the former may seek to fulfill family nutritional needs, whereas the latter may pursue muscle gain and fitness objectives. By introducing role-conditioned priors, our model captures deeper user intentions in context-specific scenarios, and lays a semantically grounded foundation for subsequent alignment-based modules.

2) Motivation Inference via Semantic Co-occurrence Reasoning: This module aims to infer users' latent motivations based on their historical interaction sequences. The core modeling principle is a semantic extension of the sequential recommendation paradigm [3]: the items a user interacts with often share semantic commonalities, and these shared aspects reflect the underlying motivational patterns behind user preferences. Therefore, we adopt a strategy termed *semantic co-occurrence aggregation*, which integrates the motivational signals embedded in each interacted item to guide the language model in performing interpretable motivation inference. The inference process is formulated as follows:

$$P(m \mid \mathcal{I}_u) \propto \sum_{i=1}^{|\mathcal{I}_u|} \omega_i \cdot P(m \mid I_i) \tag{2}$$

where $\mathcal{I}_u = \{I_1, I_2, ..., I_n\}$ denotes the set of items interacted by user u; I_i refers to the i-th item, with its semantic content encoded from textual descriptions; $P(m \mid I_i)$ measures the relevance of item I_i to motivation m; ω_i is a semantic weight assigned to I_i, which can be computed via item co-occurrence statistics, TF-IDF, or embedding similarity; $P(m \mid \mathcal{I}_u)$ is the aggregated motivation distribution inferred for the user.

This formulation reflects a key modeling assumption: *user motivation emerges from the collaborative aggregation of semantic cues across interacted items*. In the absence of explicit supervision, such a co-occurrence-based inference mechanism provides semantic grounding for uncovering motivations and naturally extends the classical intuition that "similar items imply similar preferences" into the motivational space. The final inference is executed via a language model as:

$$F_{\text{seq}}(\mathcal{I}_u) = \text{LLM}(\text{Prompt}_{\text{seq}}(\mathcal{I}_u)) \rightarrow M_u^{(2)} \tag{3}$$

where $\text{Prompt}_{\text{seq}}(\mathcal{I}_u)$ denotes a sequence-based prompt composed of item semantics, and $M_u^{(2)}$ represents the candidate motivation list generated in this stage. This semantic co-occurrence reasoning mechanism enables bottom-up motivation inference by capturing implicit correlations among

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

interacted items, offering a content-driven collaborative perspective to interpret the user's underlying behavioral drivers.

3) Motivation Fusion via Conflict-Aware Reasoning: This module aims to integrate the motivation sets derived from the two preceding reasoning pathways into a more consistent and semantically coherent representation. We define this process as a conflict-aware fusion function implemented by an LLM:

$$M_u = F_{\text{merge}}\left(M_u^{(1)}, M_u^{(2)}\right) \tag{4}$$

Here, $M_u^{(1)}$ denotes the set of motivations inferred from the user's social role, and $M_u^{(2)}$ represents the motivations derived from the semantic aggregation of interacted items. The output M_u is the final, unified motivation set.

The function $F_{\text{merge}}(\cdot)$ is executed via a prompt-driven inference process within the LLM. It takes the two input motivation sets and performs semantic conflict detection, followed by a fusion operation that resolves contradictions and consolidates consistent signals. The resulting output maintains high semantic alignment and reflects an integrated representation of the user's intent.

4) Language-based Alignment Feedback (LAF) for Profile: We propose a novel alignment paradigm in recommender systems as a core innovation in the DAMA framework. Unlike conventional alignment methods in reinforcement learning (e.g., RLHF or DPO), which typically rely on numerical scores [17] or pairwise preference comparisons [18] to optimize model outputs, we propose to utilize real user-generated textual feedback—such as reviews, affective expressions, and linguistic evaluations—as a more fine-grained and interpretable form of natural language supervision to achieve semantic alignment in motivation modeling.

a) User Feedback Dataset Construction: For a user u with a historical interaction sequence $H_u = \{i_k\}_{k=1}^{K}$, we construct a personalized feedback dataset as follows:

$$D_u = \{(i_k, r_k)\}_{k=1}^{K} \tag{5}$$

Here, i_k denotes the k-th item the user interacted with, and r_k refers to the associated subjective feedback (e.g., reviews, affective tags, or textual comments). Compared with traditional rating scores, r_k provides richer insight into human intent and authentic motivational drivers.

b) Motivation Alignment Optimization via Language Feedback: After obtaining the initial motivation set M_u, we define the following alignment objective:

$$\mathcal{L}_{\text{LAF}}(M_u', D_u) = \mathcal{L}_{\text{div}}(M_u', D_u) + \lambda \cdot \mathcal{L}_{\text{cons}}(M_u', M_u) \tag{6}$$

where: M_u' denotes the motivation set updated via linguistic feedback; $\mathcal{L}_{\text{div}}(M_u', D_u)$ measures the semantic divergence between the updated motivations and the language feedback; $\mathcal{L}_{\text{cons}}(M_u', M_u^{\text{init}})$ quantifies the consistency between the updated and initial motivation sets; λ is a tunable hyperparameter balancing alignment fidelity and reasoning continuity.

This definition reflects our core philosophy: to guide the updated motivation set M_u' toward better alignment with the user's true intentions. By integrating both behavioral signals (what the user *did*) and linguistic feedback (what the user *said*), we provide the language model with a semantically grounded alignment signal. This joint modeling ensures that the updated motivations reflect authentic user feedback while maintaining continuity with prior inferences.

c) Reflective Update via Language Models: Finally, we guide the motivation update through an LLM-based reflective reasoning process:

$$M_u^{\text{final}} = \text{LLM}(\text{Prompt}_{\text{reflect}}(M_u, D_u)) \tag{7}$$

That is, we compose a carefully designed prompt combining the initial motivation set and user feedback, prompting the language model to conduct reflective semantic reasoning toward the above alignment objective to produce the final, semantically aligned motivation set M_u^{final}.

Notably, this language-driven alignment paradigm (LAF) is, to our knowledge, the first proposed for recommender systems. By leveraging natural language feedback, LAF offers a more human-centric and interpretable alignment mechanism. Unlike reward-based methods, it preserves affective signals and contextual nuances, enabling high-fidelity alignment with user intent through a structured, semantics-driven update process.

B. Experience-Aligned Attribute Extraction Module

1) Semantic Abstraction from Product Descriptions: The initial objective is to extract a structured set of user-perceivable attributes from the natural language descriptions of candidate products, serving to capture their core functionalities and potential appeal. This process is formulated as a joint optimization problem that balances semantic abstraction with motivational constraints:

$$A_p^{\text{init}} = \arg\min_{A} \left[\mathcal{L}_{\text{recon}}(d_p, A) + \lambda \cdot \mathcal{L}_{\text{align}}(A, M_u)\right] \tag{8}$$

where: d_p: the natural language description of product p; A_p^{init}: the initial attribute set extracted from d_p; M_u: the motivation of user u, obtained from the preceding module; $\mathcal{L}_{\text{recon}}(d_p, A)$: the reconstruction loss measuring how well the attributes preserve the semantic content of d_p; $\mathcal{L}_{\text{align}}(A, M_u)$: the compatibility loss evaluating the semantic coherence between the attributes and the user's motivational intent (serving as a soft constraint rather than a strict alignment target); λ: a trade-off coefficient balancing semantic fidelity and motivational relevance.

This formulation serves two main purposes: it enables semantic abstraction by compressing unstructured product descriptions into computable attribute representations, thereby enhancing perceptual efficiency and generalization capacity. Simultaneously, the motivation acts as a soft regularization signal, guiding the abstraction process toward user-relevant attribute dimensions and improving personalization.

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

2) Language-based Alignment Feedback (LAF) for Attribute: To enhance the alignment between attribute modeling and users' perceived experiences, we extend the Language-based Alignment Feedback (LAF) paradigm from motivation modeling to product-side attribute extraction, serving as a unified alignment principle within the DAMA framework. Unlike conventional abstraction methods that rely solely on product-side descriptions, this module incorporates real user-generated textual feedback—such as reviews, affective expressions, and linguistic evaluations—as natural language supervision signals to optimize attribute representations toward semantic consistency with user experience.

a) User Experience Dataset Construction: For each candidate product p, a personalized feedback dataset is constructed based on its historical user experience feedback:

$$R_p = \{(u_i, r_i)\}_{i=1}^{N} \tag{9}$$

Here, u_i denotes a user who has interacted with product p, and r_i is the corresponding subjective feedback (e.g., review, experiential comment, or affective expression). Compared with the product's own description, such feedback captures user-centered perceptions of product attributes, helping correct potential bias, omission, or marketing exaggeration in the original descriptions.

b) Attribute Alignment Optimization via Language Feedback: Given the initial attribute set A_p^{init} extracted from the product description, we define the following alignment objective:

$$\mathcal{L}_{\text{LAF}}(A'_p, R_p) = \mathcal{L}_{\text{div}}(A'_p, R_p) + \lambda \cdot \mathcal{L}_{\text{cons}}(A'_p, A_p^{\text{init}}) \tag{10}$$

Here, A'_p denotes the updated attribute set aligned via user feedback; $\mathcal{L}_{\text{div}}(A'_p, R_p)$ measures the semantic divergence between the revised attributes and user reviews; $\mathcal{L}_{\text{cons}}(A'_p, A_p^{\text{init}})$ quantifies the consistency between the updated and initial attribute representations; λ is a tunable hyperparameter balancing the trade-off between alignment fidelity and preservation of original semantic abstraction.

This formulation reflects the unified alignment philosophy: instead of overriding the initial abstraction, it refines it in light of user-centered semantic evidence. By jointly considering "how the product is described" and "how users actually experience it," this alignment process embeds user-grounded reasoning into the attribute modeling pipeline.

c) Reflective Update via Language Models: Finally, we employ a large language model (LLM) to perform reflective reasoning based on the user feedback:

$$A_p^{\text{final}} = \text{LLM}(\text{Prompt}_{\text{reflect}}(A_p^{\text{init}}, R_p)) \tag{11}$$

That is, we compose a carefully designed prompt combining the initial attribute set and user feedback, prompting the LLM to conduct reflective semantic reasoning toward the above alignment objective and produce the final, semantically aligned attribute set A_p^{final}.

Notably, this is the first attempt to apply LAF to product-side attribute modeling in recommender systems. It shares a consistent principle, objective, and modeling paradigm with the LAF module for user motivation, collectively forming a unified language-driven alignment approach under the DAMA framework.

C. Alignment-Guided Recommendation Module

This module aims to generate the final recommendation list by aligning the user motivation vector M_u with the attribute sets $\{A_p\}_{p=1}^{P}$ of candidate products. To mitigate potential hallucinations or misjudgments from large language models (LLMs) in one-shot decisions, we adopt a multi-cycle filtering strategy guided by alignment constraints.

We first define a motivation alignment scoring function:

$$S(u, p) = \text{sim}(M_u, A_p) \tag{12}$$

where u denotes the user; p denotes a candidate product; M_u is the motivation vector representing user u's internal goals and preferences; A_p is the set of semantic attributes extracted for product p; $\text{sim}(M_u, A_p)$ computes the semantic compatibility between the user's motivation and product's attributes; $S(u, p)$ is the scalar alignment score between user u and product p.

Instead of ranking all candidates in a single pass, we iteratively eliminate the least aligned items in a cycle-wise manner. At each round t, we define a candidate pool $C^{(t)} \subseteq \{1, 2, \ldots, P\}$, and update it by removing the bottom δ_t candidates with the lowest alignment scores:

$$C^{(t+1)} = C^{(t)} \setminus \text{Bottom}_{\delta_t}(S(u, p)), \quad \text{for } p \in C^{(t)} \tag{13}$$

where $C^{(t)}$ is the set of remaining candidate product indices at round t; δ_t is the number of candidates to discard at round t; $\text{Bottom}_{\delta_t}(S(u, p))$ returns the δ_t candidates with the lowest alignment scores in $C^{(t)}$.

This iterative selection procedure constitutes a local confidence optimization under motivation-alignment constraints, whereby the system progressively retains items with higher semantic compatibility while filtering out semantically irrelevant or misaligned candidates. By avoiding brittle one-shot decisions, the model gains robustness against noisy alignment scores and achieves more reliable user preference reflection.

V. Experiment

A. *Experiment setting*

To thoroughly evaluate our proposed framework, we use three benchmark datasets from the 2018 Amazon dataset [19], namely *Beauty*, *Toys*, and *Sports*. Table I summarizes the number of users, products, and interactions in each dataset. Each interaction includes reviewerID, asin(product ID), rating, and product-specific textual information.

To ensure sufficient data for inference, we used the 5-core version of each dataset, meaning that all users have more than 5 interactions. For each user, we treated the items with a rating greater than 3 in their interaction history as preferred products. Then, to construct the test set, we randomly sampled 29 negative items, and selected the last positively rated item

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

(rating $>$ 3) in the user's sequence as the positive item, resulting in a candidate set of 30 items in total.

TABLE I: User, Item, and Interaction Data

Category	#User	#Item	#Interaction	#Sparsity
Beauty	22269	259204	157373	99.9972%
Sports	35553	532197	250024	99.9986%
Toys	19377	336072	134866	99.9979%

We evaluate performance using two standard metrics: Hit Rate@K (HR@K) and Normalized Discounted Cumulative Gain@K (NDCG@K). These metrics are calculated for each user and then averaged across all users. Following common practice, we use K=5 and K=10 to assess performance at different recommendation depths.

For the implementation details, our model uses GPT-3.5-turbo with default hyperparameters. The maximum number of product feedback is set to 50, and the dataset size is limited to 500 samples.

For baseline selection, we categorize the baselines into three types. The first type includes traditional collaborative filtering–based recommendation systems, such as NCF [20], BERT4Rec [21], GRU4Rec [2], and SASRec [3]. The second type consists of models that perform recommendation using purely textual information, including TallRec [12], P5 [11], and RDRec [14]. The third type contains models that integrate LLMs with traditional recommendation systems for multimodal alignment, such as A-LLMRec [16].

B. *Overall performance*

We first compared the overall performance differences across models. As shown in the Table II, the results clearly illustrate substantial performance gaps between models. Next, we provide a detailed analysis of these differences from multiple perspectives:

Overall Performance Comparison: Our proposed method significantly outperforms existing recommendation models on all key metrics—Hit Rate and NDCG—across the Beauty, Sports, and Toys datasets, demonstrating a clear overall advantage. Specifically, on all mainstream evaluation metrics (Hit@5/10, NDCG@5/10), our method not only leads in absolute scores but also shows substantial improvements. For instance, on the Toys dataset, NDCG@10 improves by approximately 30% compared to A-LLMRec, and by over 70% compared to traditional collaborative filtering (CF) methods. These results strongly validate our LLM alignment–based framework, which effectively enriches user interaction data, deeply matches product features with user motivations, and significantly enhances the model's ability to represent the user–item relationship.

Comparison with Traditional Collaborative Filtering (CF) Models: Traditional CF models (e.g., NCF, GRU4Rec, SASRec, BERT4Rec) typically rely solely on historical user–item interactions to infer user preferences and product features. However, these models perform poorly when user or item interactions are sparse, as they ignore additional information embedded in the users or products themselves. In contrast, our model not only focuses on user–item interaction sequences but also explores intrinsic product information and user-provided feedback. This modeling approach captures user and product characteristics more comprehensively and enables more accurate matches between the two.

Comparison with Text-Based Recommendation Models: Recent LLM-based models like TallRec and P5 rely primarily on textual information for recommendation. However, these models tend to represent product information too simplistically. For example, P5 treats product information as ID tokens in a traditional recommendation setup, and TallRec uses only the product name to encode product features. In real-world scenarios, user purchase decisions are rarely based solely on product names or IDs—they often depend more on detailed product descriptions and feedback from other customers. Although RDRec takes a step further by extracting user preferences and product features from feedback using a large model and modeling their relationships with a smaller model, it lacks clear guidance. The model must learn the relationships independently, making it difficult to capture the deeper connection between users and products accurately and completely. In contrast, our model integrates richer information sources, starting from authentic user feedback and explicitly models the relationship between product features and user motivations. This ensures that the LLM can more precisely extract and match deep semantic relationships between users and products.

Comparison with Multimodal Alignment–Based Recommendation Models: A-LLMRec incorporates the outputs of traditional collaborative filtering models as soft prompts fed into the LLM, aiming to improve recommendation by fusing multimodal information. However, this method mainly emphasizes simple fusion between modalities. In contrast, our method focuses on extracting authentic user feedback and capturing real purchase motivations and inherent product characteristics in an explicit and accurate way. Our proposed alignment strategy emphasizes a clear and truthful correspondence between user feedback and product features, ensuring that the model's recommendations are not only accurate but also interpretable.

C. *Cold start*

In this section, we primarily analyze the performance of different models under cold start scenarios (Table III). In the experimental design, we further divide the cold start into two types: soft cold start and hard cold start.

Specifically, soft cold start involves selecting the bottom 10% of users or items (based on interaction frequency) as the test set after the model has been trained on the full dataset. This setup aims to evaluate the model's recommendation ability for users or items with limited interaction records—in other words, its generalization ability to "weakly-interacted" users or items. In contrast, hard cold start is a more stringent condition, where the bottom 10% of users or items are completely excluded from the training set. This tests the model's performance on entirely unseen users or items.

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

TABLE II: This table shows the experimental results of ours model compared with all baseline models. To ensure the accuracy of the experiments, we conducted three performance evaluations for each model and reported their average values.

Dataset	Metric	NCF	GRU4Rec	SASRec	BERT4Rec	Tallrec	P5	A-LLM4Rec	RDRec	DAMA
Beauty	HT@5	0.469	0.3244	0.3095	0.3007	0.32	0.2448	0.5307	0.2651	**0.602**
	HT@10	0.62	0.5101	0.5016	0.4842	0.5252	0.3441	0.5365	0.3451	**0.744**
	NDCG@5	0.341	0.21	0.197	0.1885	0.2097	0.1673	0.339	0.1966	**0.4713**
	NDCG@10	0.39	0.271	0.2619	0.2473	0.255	0.1993	0.358	0.2219	**0.5176**
Sports	HT@5	0.408	0.2566	0.2744	0.275	0.35	0.2241	0.218	0.2619	**0.648**
	HT@10	0.588	0.4475	0.4647	0.4611	0.58	0.3313	0.3546	0.3472	**0.794**
	NDCG@5	0.285	0.1481	0.1732	0.174	0.2122	0.1483	0.1394	0.2011	**0.5091**
	NDCG@10	0.336	0.2097	0.2344	0.2366	0.2907	0.1827	0.2	0.2295	**0.5558**
Toys	HT@5	0.413	0.2899	0.2759	0.2816	0.3961	0.1411	0.321	0.1673	**0.708**
	HT@10	0.576	0.4875	0.4616	0.4744	0.6091	0.2227	0.436	0.241	**0.788**
	NDCG@5	0.295	0.1764	0.1738	0.1775	0.2659	0.0916	0.2496	0.1178	**0.5814**
	NDCG@10	0.342	0.239	0.2334	0.2397	0.3271	0.1178	0.3027	0.141	**0.6072**

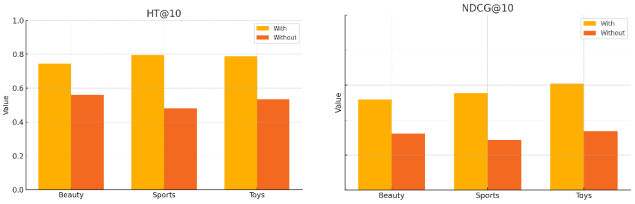
From the overall experimental results, traditional collaborative filtering models generally perform better in the soft cold start scenario than in the hard cold start scenario. This is because such models heavily rely on whether the user or item has been seen during training. In other words, they model known users much better than new users. In contrast, LLM-based models demonstrate more stable performance across both scenarios. This is due to the fact that LLM models rely not only on user-item interaction sequences but also on the semantic information embedded in item descriptions. Even if an item has never appeared in the training phase, the model can extract semantic features from its textual description and thereby make reasonable recommendations. However, our model further enhances this capability by introducing a dual-perspective user–item alignment mechanism on top of LLMs, allowing it to capture richer and more precise motivational features, especially in cold-start scenarios.

Further analyzing the performance differences between the item cold start and user cold start scenarios, we observe that all model types generally perform worse in the item cold start condition than in the user cold start. This phenomenon is particularly pronounced in traditional collaborative filtering models, mainly due to their ID-based training mechanism. If a product appears infrequently in the training set, its corresponding embedding is updated less often, making it difficult for the model to learn accurate representations. During inference, these models tend to favor frequently seen items, which results in a lower recommendation accuracy for cold-start items. In the hard cold start scenario, where cold-start items are completely absent from training data, the model can hardly establish any representation, leading to a significant drop in performance. In contrast, our model benefits from the LLM's ability to utilize textual data. Even when samples are sparse, it can extract valuable semantic features from item descriptions. Additionally, by enriching the content associated with each item, we further enhance the model's ability to infer meaningful user and item features under conditions of limited information.

In conclusion, experimental results demonstrate that our proposed model exhibits strong robustness and generalization across various cold start scenarios, validating its effectiveness and resilience in recommendation tasks under data-scarce conditions.

D. Ablation study

In this section, we focus on the specific impact of the Experience-Aligned Attribute Extraction Module (Fig. 2), multi-round in Alignment-Guided Recommendation Module (Fig. 3), LLM in Alignment-Guided Recommendation Module (Fig. 4) and feedback number (Fig. 5) on the overall performance of the model.



(a) HT result w/o Experience-Aligned Attribute Extraction module

(b) NDCG result w/o Experience-Aligned Attribute Extraction module

Fig. 2: Ablation results of removing the Experience-Aligned Attribute Extraction module from the DAMA.

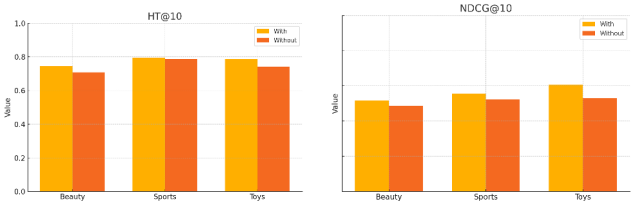
Impact of Experience-Aligned Attribute Extraction Module on Performance: We conducted comparative experiments with and without the Experience-Aligned Attribute Extraction module on the Beauty, Sports, and Toys datasets, using HT@10 and NDCG@10 as evaluation metrics. From the bar chart of the results, it is clear that removing the Experience-Aligned Attribute Extraction module leads to a significant performance drop—the overall performance drops to about two-thirds of the original. This highlights the importance of the Experience-Aligned Attribute Extraction module, as it directly ensures whether the model can accurately match product features with user real motivations. Without this module, the LLM suffers in its ability to align core product features, and thus cannot

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

TABLE III: Performance on Various Cold Start Settings

Setting	Dataset	NCF	GRU4Rec	SASRec	BERT4Rec	Tallrec	DAMA
Hard Cold Start Product							
	Beauty HT@10	0.019	0.0073	0.0102	0.0304	0.507	**0.68**
	Beauty NDCG@10	0.006	0.0023	0.0035	0.01	0.2309	**0.4027**
	Sports HT@10	0.015	0.0070	0.017	0.0093	0.5016	**0.68**
	Sports NDCG@10	0.005	0.0003	0.0051	0.0042	0.2333	**0.3999**
	Toys HT@10	0.112	0.0258	0.0121	0.0271	0.5258	**0.735**
	Toys NDCG@10	0.037	0.0115	0.0086	0.0114	0.2508	**0.4802**
Hard Cold Start User							
	Beauty HT@10	0.557	0.4087	0.5096	0.5163	0.52	**0.7**
	Beauty NDCG@10	0.323	0.2513	0.2657	0.265	0.251	**0.462**
	Sports HT@10	0.588	0.4246	0.4754	0.4587	0.5603	**0.77**
	Sports NDCG@10	0.337	0.1841	0.2573	0.2199	0.2704	**0.5143**
	Toys HT@10	0.532	0.4692	0.5137	0.4863	0.5871	**0.78**
	Toys NDCG@10	0.299	0.2249	0.2639	0.239	0.3048	**0.6162**
Soft Cold Start Product							
	Beauty HT@10	0.329	0.1384	0.0211	0.0408	0.5	**0.68**
	Beauty NDCG@10	0.157	0.0627	0.0076	0.0142	0.2317	**0.4027**
	Sports HT@10	0.325	0.211	0.1034	0.087	0.513	**0.68**
	Sports NDCG@10	0.143	0.1151	0.0373	0.0311	0.2373	**0.3999**
	Toys HT@10	0.415	0.1503	0.0413	0.0508	0.539	**0.735**
	Toys NDCG@10	0.218	0.0681	0.0143	0.0182	0.2873	**0.4802**
Soft Cold Start User							
	Beauty HT@10	0.574	0.5113	0.5263	0.5088	0.5217	**0.7**
	Beauty NDCG@10	0.343	0.2746	0.2747	0.2607	0.2505	**0.462**
	Sports HT@10	0.581	0.4445	0.4787	0.4696	0.57	**0.77**
	Sports NDCG@10	0.332	0.2024	0.255	0.2405	0.2711	**0.5143**
	Toys HT@10	0.549	0.4815	0.4976	0.4927	0.5912	**0.78**
	Toys NDCG@10	0.311	0.2387	0.2569	0.2435	0.3177	**0.6162**

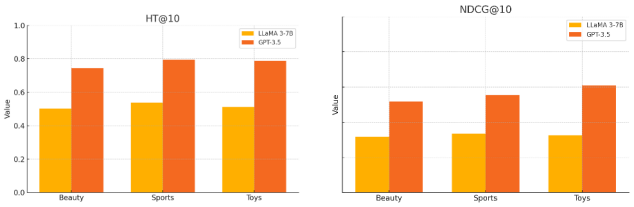
accurately understand and establish the relationship between features and motivations.



(a) HT result w/o multi-round (b) NDCG result w/o multi-round

Fig. 3: Ablation results of removing the multi-round from the Alignment-Guided Recommendation Module.

Impact of multi round(Alignment-Guided Recommendation Module) on Performance: Multi-round is a strategy we proposed to alleviate the hallucination issue in LLMs in Alignment-Guided Recommendation Module. If the LLM is asked to select the top-k candidate products in one shot, it may hallucinate and mistakenly filter out the true positive items. Therefore, we proposed a multi-round approach where only a small number of low-ranked items are eliminated in each round, gradually approaching the final target number k. Experimental results clearly show a significant performance improvement when using the multi-round method. This result fully demonstrates the effectiveness of the strategy and confirms its ability to mitigate performance degradation caused by hallucinations.



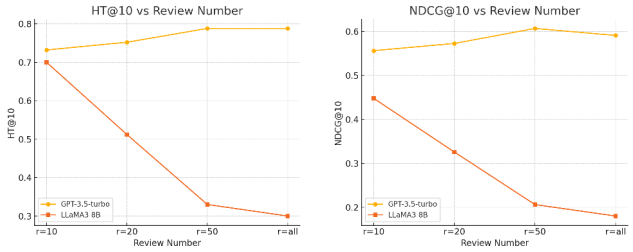
(a) HT for different LLM in Alignment-Guided Recommendation Module (b) NDCG for different LLM in Alignment-Guided Recommendation Module

Fig. 4: The effectiveness of different LLM in Alignment-Guided Recommendation Module.

Impact of Different LLM in Alignment-Guided Recommendation Module on Performance: The Alignment-Guided Recommendation Module is a key module responsible for linking product features with user motivations and is crucial for precise information alignment. We compared a fine-tuned open-source model (LLaMA3 8B) with a closed-source model (GPT-3.5 Turbo). To fine-tune LLaMA3 8B, we constructed a dataset with approximately 3,000 samples, each containing specific product information along with true (positive) or false (negative) feature labels. The model learns to determine whether a feature belongs to the corresponding product. Experimental results show that GPT-3.5 Turbo performs better overall. This is because, although the fine-tuned open-source

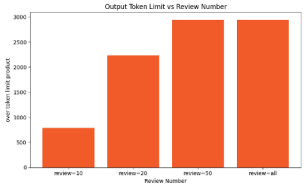
Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

model may excel in domain-specific knowledge, GPT-3.5 Turbo possesses stronger general semantic understanding and more nuanced and comprehensive feature extraction capabilities. This allows it to more effectively establish precise alignment for product features and user motivations and match the real product feature with user motivation, leading to superior overall recommendation performance.



(a) HT for difference feedback number

(b) NDCG for difference feedback number



(c) Number of Products Exceeding Token Limit

Fig. 5: The effectiveness of different feedback number.

Impact of feedback Quantity on Model Performance: In this section, we investigate the effect of the number of product feedback on the overall performance of the model. We selected four settings with different feedback quantities: 10, 20, 50, and all available feedback, and analyzed how these variations affect overall performance. Additionally, we compared the performance trends of an open-source LLM (LLaMA3 8B) and a closed-source LLM (GPT-3.5 Turbo) under these conditions.

From the experimental results, it is evident that for the closed-source LLM, model performance improves gradually as the number of feedback increases. However, this improvement tends to stabilize once the feedback count reaches 50. This indicates that with 50 feedback, the model can already extract sufficient product features from user feedback. Moreover, the overall performance curve is not particularly steep, suggesting that as long as genuine user feedback are present, the model can effectively identify key product characteristics. In contrast, relying solely on product self-descriptions is often insufficient, as commercial descriptions tend to contain marketing bias.

In comparison, the performance trend for the open-source LLM is the opposite—it declines as the number of feedback increases. This is primarily due to the limited token capacity of the open-source model (set to 2048 tokens in our experiments). When the number of feedback is too large, input truncation due to token overflow can occur, leading to incomplete information being fed into the model, thereby reducing its prediction accuracy.

To verify this hypothesis, we counted the number of products that exceeded the token limit at different feedback thresholds and visualized the results (as shown in the Fig. 5c). The chart clearly shows that as the number of feedback increases, so does the number of products surpassing the token limit, which in turn causes the model performance to deteriorate.

In future work, we aim to explore how to enhance the semantic understanding capabilities of open-source models under strict token constraints.

E. Further Discussion on Hallucination

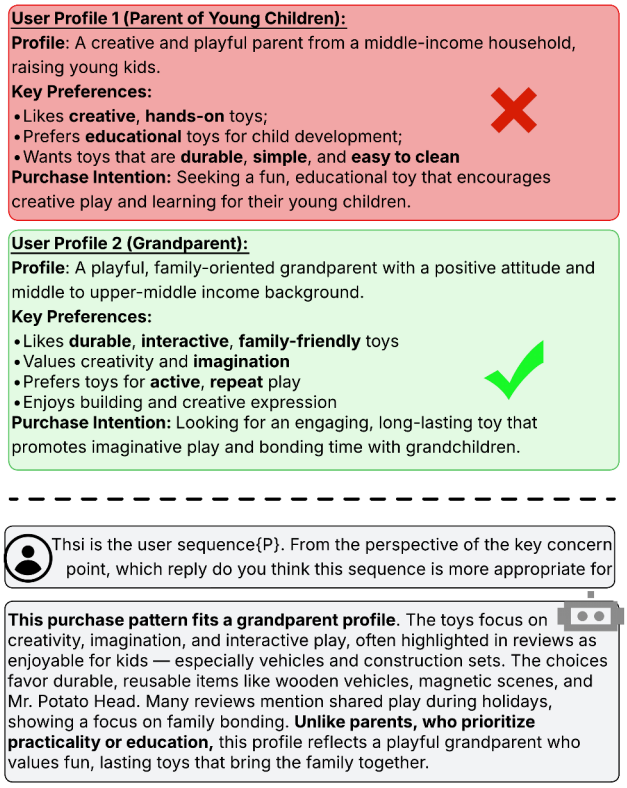


Fig. 6: Comparison of LLM responses given different levels of context: Above the dashed line are two different input profiles; below the dashed line is a comparison of GPT's responses to the two profiles and the actual user interaction sequence.

In this section, we present a representative example of a user's interaction history to illustrate how large language models (LLMs) may generate hallucinated content when provided with incomplete contextual information and how the Language-based Alignment Feedback(LAF) paradigm can align user's real motivation. As depicted in Fig. 6, the area above the dashed line contrasts two experimental conditions: the red panel displays the model's output when only product titles from the user's interaction history are available, while the green panel illustrates the output generated under a more informative condition, wherein the model is supplied with a richer and more detailed representation that includes expanded product attributes.

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.

Subsequently, both sets of outputs—along with the full user interaction history containing complete product information—were fed into a new instance of ChatGPT. The task was to assess which output better aligned with the user's genuine preferences. According to ChatGPT's judgment, the output generated using more complete product information was significantly more consistent with the user's actual interests. Conversely, the output based solely on product titles exhibited clear signs of hallucination: the model erroneously inferred a user preference for certain attributes (e.g., products with strong educational value), despite no such preferences being expressed in the user's actual purchasing behavior or feedback.

This experiment demonstrates that when contextual information is insufficient, LLMs are highly susceptible to hallucinations and overconfidence, which may lead to compounding errors in downstream reasoning and adversely affect the accuracy of recommendation results. Therefore, providing structured and information-rich context is critical for ensuring the robustness and reliability of LLM-based recommendation systems.

VI. Conclusion

This paper presents DAMA, a motivation-aware recommendation framework that tackles hallucination and overconfidence in LLM-powered recommendation systems through a dual alignment mechanism. Extensive experiments demonstrate its effectiveness, particularly in cold start scenarios. The proposed language-based alignment feedback (LAF) offers a novel, human-centric strategy to refine model outputs via natural language guidance. Our results highlight LLM alignment as a promising direction for building trustworthy and interpretable recommendation systems.

References

[1] J. Bobadilla, F. Ortega, A. Hernando, and A. Gutiérrez, "Recommender systems survey," *Knowledge-Based Systems*, vol. 46, pp. 109–132, 2013.

[2] B. Hidasi, A. Karatzoglou, L. Baltrunas, and D. Tikk, "Session-based recommendations with recurrent neural networks," 2016. [Online]. Available: https://arxiv.org/abs/1511.06939

[3] W.-C. Kang and J. McAuley, "Self-attentive sequential recommendation," 2018. [Online]. Available: https://arxiv.org/abs/1808.09781

[4] S. Lam, D. Frankowski, and J. Riedl, "Do you trust your recommendations? an exploration of security and privacy issues in recommender systems," 01 2006, pp. 14–29.

[5] L. Wu, Z. Zheng, Z. Qiu, H. Wang, H. Gu, T. Shen, C. Qin, C. Zhu, H. Zhu, Q. Liu, H. Xiong, and E. Chen, "A survey on large language models for recommendation," 2024. [Online]. Available: https://arxiv.org/abs/2305.19860

[6] J. Wang, H. Lu, J. Caverlee, E. Chi, and M. Chen, "Large language models as data augmenters for cold-start item recommendation," 2024. [Online]. Available: https://arxiv.org/abs/2402.11724

[7] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, Mar. 2023. [Online]. Available: https://doi.org/10.1145/3571730

[8] Y. Xiao and W. Y. Wang, "On hallucination and predictive uncertainty in conditional language generation," in *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, P. Merlo, J. Tiedemann, and R. Tsarfaty, Eds. Online: Association for Computational Linguistics, Apr. 2021, pp. 2734–2744. [Online]. Available: https://aclanthology.org/2021.eacl-main.236/

[9] J. Liu, C. Liu, P. Zhou, Q. Ye, D. Chong, K. Zhou, Y. Xie, Y. Cao, S. Wang, C. You, and P. S. Yu, "Llmrec: Benchmarking large language models on recommendation task," 2023. [Online]. Available: https://arxiv.org/abs/2308.12241

[10] Z. Zhao, W. Fan, J. Li, Y. Liu, X. Mei, Y. Wang, Z. Wen, F. Wang, X. Zhao, J. Tang, and Q. Li, "Recommender systems in the era of large language models (llms)," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 11, pp. 6889–6907, 2024.

[11] S. Geng, S. Liu, Z. Fu, Y. Ge, and Y. Zhang, "Recommendation as language processing (rlp): A unified pretrain, personalized prompt & predict paradigm (p5)," in *Proceedings of the 16th ACM Conference on Recommender Systems*, ser. RecSys '22. New York, NY, USA: Association for Computing Machinery, 2022, p. 299–315. [Online]. Available: https://doi.org/10.1145/3523227.3546767

[12] K. Bao, J. Zhang, Y. Zhang, W. Wang, F. Feng, and X. He, "Tallrec: An effective and efficient tuning framework to align large language model with recommendation," in *Proceedings of the 17th ACM Conference on Recommender Systems*, ser. RecSys '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 1007–1014. [Online]. Available: https://doi.org/10.1145/3604915.3608857

[13] H. Lyu, S. Jiang, H. Zeng, Y. Xia, Q. Wang, S. Zhang, R. Chen, C. Leung, J. Tang, and J. Luo, "Llm-rec: Personalized recommendation via prompting large language models," 2024. [Online]. Available: https://arxiv.org/abs/2307.15780

[14] X. Wang, J. Cui, Y. Suzuki, and F. Fukumoto, "RDRec: Rationale distillation for LLM-based recommendation," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, L.-W. Ku, A. Martins, and V. Srikumar, Eds. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 65–74. [Online]. Available: https://aclanthology.org/2024.acl-short.6/

[15] Y. Zhao, J. Wu, X. Wang, W. Tang, D. Wang, and M. de Rijke, "Let me do it for you: Towards llm empowered recommendation via tool learning," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, ser. SIGIR '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 1796–1806. [Online]. Available: https://doi.org/10.1145/3626772.3657828

[16] S. Kim, H. Kang, S. Choi, D. Kim, M. Yang, and C. Park, "Large language models meet collaborative filtering: An efficient all-round llm-based recommender system," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ser. KDD '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 1395–1406. [Online]. Available: https://doi.org/10.1145/3637528.3671931

[17] Z. Wang, B. Bi, S. K. Pentyala, K. Ramnath, S. Chaudhuri, S. Mehrotra, Zixu, Zhu, X.-B. Mao, S. Asur, Na, and Cheng, "A comprehensive survey of llm alignment techniques: Rlhf, rlaif, ppo, dpo and more," 2024. [Online]. Available: https://arxiv.org/abs/2407.16216

[18] T. Shen, R. Jin, Y. Huang, C. Liu, W. Dong, Z. Guo, X. Wu, Y. Liu, and D. Xiong, "Large language model alignment: A survey," 2023. [Online]. Available: https://arxiv.org/abs/2309.15025

[19] J. Ni, J. Li, and J. McAuley, "Justifying recommendations using distantly-labeled reviews and fine-grained aspects," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, K. Inui, J. Jiang, V. Ng, and X. Wan, Eds. Hong Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 188–197. [Online]. Available: https://aclanthology.org/D19-1018/

[20] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua, "Neural collaborative filtering," 2017. [Online]. Available: https://arxiv.org/abs/1708.05031

[21] F. Sun, J. Liu, J. Wu, C. Pei, X. Lin, W. Ou, and P. Jiang, "Bert4rec: Sequential recommendation with bidirectional encoder representations from transformer," in *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, ser. CIKM '19. New York, NY, USA: Association for Computing Machinery, 2019, p. 1441–1450. [Online]. Available: https://doi.org/10.1145/3357384.3357895

Authorized licensed use limited to: Hong Kong University of Science and Technology. Downloaded on September 07,2026 at 07:30:16 UTC from IEEE Xplore. Restrictions apply.